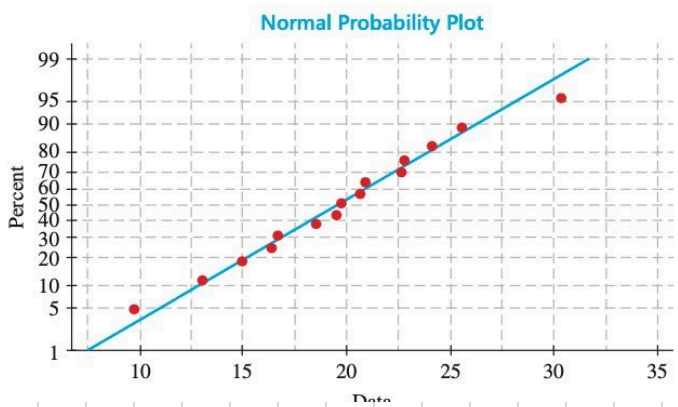
Is the Data Normal? *Use the graphical displays in Exercises 12–15 to decide whether the data have been selected from a normal population. Explain your answer.*

13.



Yes, it is a normal distribution.
y-axis is the cumulative distribution function (CDF), since most plots are concentrated around the 50% area, it follows a somewhat normal distribution.

20. Cerebral Blood Flow Cerebral blood flow (CBF) in the brains of healthy people is normally distributed with a mean of 74. A random sample of 25 stroke patients resulted in an average CBF of $\bar{x} = 69.7$ with a standard deviation of $s = 16$. Test the hypothesis that the average CBF for stroke patients is different from that of healthy people using $\alpha = .05$.

$\mu = 74$ $\quad$ $\mu \neq 74$
$n = 25$
$\bar{x} = 69.7$
$s = 16$
$\alpha = 0.05$

1) $H_0: \mu = 74$
$H_1: \mu \neq 74$

2) $a = 0.05$

3) $t_{STAT} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t(n-1)$

4) $t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{69.7 - 74}{16/\sqrt{25}}$

$= -1.34375$

5) $(25-1)$
$(0.05 \div 2)$

$t_{0.025} = \pm 2.064$

6) $\therefore$ do not reject H_0 since it lies within the critical values

p-value
$p = 0.192$ from GC

since $p = 0.192 > 0.05$, do not reject H_0

30. Bulimia An article in the *British Journal of Clinical Psychology* indicates that self-esteem was an important predictor of those who will benefit from treatment for bulimia.[4] The table gives the mean and standard deviation of self-esteem scores prior to treatment, at posttreatment, and during a follow-up:

	Pretreatment	Posttreatment	Follow-up
Sample Mean $\bar{x}$	20.3	26.6	27.7
Standard Deviation s	5.0	7.4	8.2
Sample Size n	21	21	20

a. Use a test of hypothesis to determine whether there is sufficient evidence to conclude that the true pretreatment mean is less than 25.

b. Construct a 95% confidence interval for the true posttreatment mean.

c. In Section 10.3, we will introduce small-sample techniques for making inferences about the difference between two population means. Without a formal statistical test, what might you conclude about the differences among the three sampled population means represented by the results in the table?

a. $\bar{x} = 20.3$
$s = 5.0$
$n = 21$

1) $H_0: \mu = 25$
$H_1: \mu < 25$

2) since not stated, assume $\alpha = 0.05$

3) $t_{STAT} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{(n-1)}$

4) $t^* = \frac{20.3 - 25}{5/\sqrt{21}}$

$= -4.30762$

$t_{crit} = -1.725$

$\alpha = 0.05$
$df = 21 - 1$
$= 20$

since $-4.31 < -1.725$,
reject H_0

b. $\bar{x} = 26.6$
$s = 7.4$
$n = 21$

confidence interval $= 26.6 \pm (t_{\frac{0.05}{2}, 21-1})(\frac{7.4}{\sqrt{21}})$
$= 26.6 \pm (2.086 \times 1.61481)$
$= 26.6 \pm 3.3685$
$= (23.232, 29.968)$

c.

35. Sodium Chloride Measurements of water intake, obtained from a sample of 17 rats that had been injected with a sodium chloride solution, produced a mean and standard deviation of 31.0 and 6.2 cubic centimeters (cm^3), respectively. Suppose that the average water intake for noninjected rats observed over a comparable period of time is 22.0 cm^3.

a. Do the data indicate that injected rats drink more water than noninjected rats? Test at the 5% level of significance.

b. Find a 90% confidence interval for the mean water intake for injected rats.

a. 1) $H_0: \mu = 22$ $\quad \bar{x} = 31$

$H_1: \mu > 22$ $\quad s = 6.2$

2) $\alpha = 0.05$

3) $t_{STAT} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

4) $t^* = \frac{31-22}{6.2/\sqrt{17}}$

$= 5.985$

5) rejection region $= t_{17-1\ 0.05}$

$= 1.746$

6) since $5.985 > 1.746$, do not reject H_1, $\therefore$ indicates that injected rats drink more

b. confidence interval $= 31 \pm t_{\frac{0.10}{2}, 17-1} \left(\frac{s}{\sqrt{n}}\right)$

$= 31 \pm \left(1.746 \times \frac{6.2}{\sqrt{17}}\right)$

$= 31 \pm 2.6255$

$= (28.375, 33.625)$

Two-Sample Tests *Use the information provided in Exercises 7–8. State the null and alternative hypotheses for testing for a significant difference in means, calculate the pooled estimate of σ^2, the associated degrees of freedom, and the observed value of the t statistic. What is the rejection region using $\alpha = .05$? What is the p-value for the test? What can you conclude from these data?*

7.

Population 1	12	3	8	5	
Population 2	14	7	7	9	6

7. $H_0: \mu_1 - \mu_2 = 0$
$H_a: \mu_1 - \mu_2 \neq 0$ two tailed test

$\mu_1 = \frac{12+3+8+5}{4}$
$= 7$

$s_1^2 = \frac{(12-7)^2 + (3-7)^2 + (8-7)^2 + (5-7)^2}{4-1}$

$= 15.33$

$\mu_2 = \frac{14+7+7+9+6}{5}$
$= 8.6$

$s_2^2 = \frac{(14-8.6)^2 + (7-8.6)^2 + \cdots}{5-1}$

$= 10.3$

$\frac{15.33}{10.3} = 1.4883 < 3$, equal variance assumption is reasonable

$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

$= \frac{(4-1)(15.33) + (5-1)(10.3)}{4+5-2}$

$= 12.46$

$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{s^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

$= \frac{7-8.6}{\sqrt{12.46\left(\frac{1}{4}+\frac{1}{5}\right)}}$

$= -0.675$

rejection region:
$t_{\frac{0.05}{2}, 4+5-2} = \pm 2.365$

since $|-0.675| < 2.365$,
fail to reject

p-value:
since $0.68 < 1.415$, probability > 0.1
since two tailed, $p > 2(0.1)$
$p > 0.2$
$\therefore p > 0.2 > 0.05$, fail to reject

8.

Population	1	2
Sample Size	16	13
Sample Mean	34.6	32.2
Sample Variance	4.8	5.9

8. $H_0: \mu_1 - \mu_2 = 0$
$H_a: \mu_1 - \mu_2 \neq 0$

$\frac{5.9}{4.8} = 1.229 < 3$, equal var assumption is reasonable

$s_p^2 = \frac{(16-1)(4.8) + (13-1)(5.9)}{(8+1)-2}$
$= 5.29$

$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

$= \frac{34.6-32.2}{\sqrt{5.29\left(\frac{1}{16}+\frac{1}{13}\right)}}$

$= 2.80$

rejection region:
$t_{\frac{0.05}{2}, 16+13-2} = \pm 2.052$
since $2.80 > 2.052$, reject

p-value:
since $2.80 > 2.771$, $p < 0.005$
since two-tailed, $p < 2(0.005)$
$p < 0.01$
since $p < 0.01 < 0.05$, reject H_0

12. Books or iPads? An experiment was conducted to compare the use of iPads versus regular textbooks in teaching algebra to two classes of middle school students.[6] To remove teacher-to-teacher variation, the same teacher taught both classes, and all teaching materials were provided by the same author and publisher. Suppose that after 1 month, 10 students were selected from each class and their scores on an algebra advancement test recorded. The summarized data follow.

	iPad	Textbook
Mean	86.4	79.7
Standard Deviation	8.95	10.7
Sample Size	10	10

a. Use the summary data to test for a significant difference in advancement scores for the two groups using $\alpha = .05$.

b. Find a 95% confidence interval for the difference in mean scores for the two groups.

c. In light of parts a and b, what can we say about using iPads versus traditional textbooks in teaching algebra at the middle school level?

a) $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$\frac{10.7^2}{8.95^2} = 1.429 < 3$

$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

$= \frac{(10-1)(8.95^2) + (10-1)(10.7^2)}{10+10-2}$

$= 97.3$

$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2(\frac{1}{n_1}+\frac{1}{n_2})}}$

$= \frac{86.4-79.7}{\sqrt{97.3(\frac{1}{10}+\frac{1}{10})}}$

$= 1.52$

$t_{\frac{0.05}{2}, 10+10-2} = 2.101$

since $1.52 < 2.101$,
do not reject H_0.

b) $CI = (\bar{X}_1 - \bar{X}_2) \pm t_{10+10-2, \frac{0.05}{2}} \sqrt{s^2(\frac{1}{n_1}+\frac{1}{n_2})}$

$= (86.4-79.7) \pm (2.101 \times \sqrt{97.3(\frac{1}{10}+\frac{1}{10})})$

$= 6.7 \pm 9.26$

$= (-2.56, 15.96)$

c) The difference is negligible